Aqsa Khalid
Lecturer
School of Computing

CS3002 Information Security



Malware Study and Prevention

Outline



- What malware are?
- How do they infect hosts?
- How do they propagate?
- How to detect them?
- How to prevent them?
- How to analyze them?

What is Malware?



- "A program that is inserted into a system, usually covertly, with the intent of compromising the confidentiality, integrity, or availability of the victim's data, applications, or operating system [OS] or otherwise annoying or disrupting the victim." (NIST – 2005)
- Also called malicious software or digital pests



What it can do?



- Steal personal files (documents, photos etc.)
- Delete files
- Corrupt files
- Cause annoyance
- Steal information (software serial numbers, passwords etc.)
- Use your computer as relay (zombie)
- etc.

Some Malware Terminology



How malware infects a system

- Virus: attaches itself to a host file, replicates itself when executed
- Worm: propagates copies of itself to other computers in network, without any user action
- Trojan horse: apparently useful program, contains additional malicious functionality
  - Scareware: users are tricked by scaring and motivated to perform some action. e.g. buying a software license
- Browser hijacker: modifies a web browser's settings without a user's permission, to (e.g.) inject unwanted advertising into the web pages

Some Malware Terminology



What actions a malware takes after infection

- Logic bomb: "explodes" when a condition occurs
- Backdoor (trapdoor): allows unauthorized access to functionality or a system, bypassing normal checks
- Spyware: used to spy on victim's activities on a system and/or for stealing sensitive information
  - Keylogger: captures keystrokes for spying typed sensitive info
- Ransomware: blocks access to data or functionality and asks for a ransom payment
- Zombie (bot): software on infected computers that can be used to launch attacks on *other machines*

Some Malware Terminology



What actions a malware takes after infection

- Rootkit: Set of hacker tools that allow attacker to maintain root-level (admin) access to a system, while hiding their presence
- Bootkit: infects the master boot record and spreads when a system is booted from the disk containing the malware.

Important Note: Most real-world malware overlaps several categories. For example, a single piece of malware might combine virus like propagation and spyware capabilities and might also be used to launch a DoS attack.

Example Scenarios



- A user comes across a free media player with a nice UI. When they download and install the app, the player works fine but their computer starts behaving strangely, and they notice that their personal files are missing. — trojan
- A company's network experiences a sudden surge in traffic, and employees report receiving numerous spam emails. The IT department investigates and discovers that a malicious program is rapidly spreading through the network, replicating itself and sending copies to other devices. — worm
- A disgruntled employee embeds a hidden code in the company's payroll software. Months later, when the employee is fired, the code activates, deleting all employee records and causing significant disruption to the company's operations. — logic bomb

Example Scenarios



- A user's files become inaccessible, and they see a message demanding a payment in exchange for the decryption of their locked files. — ransomware
- A hacker gains access to a company's servers and installs a hidden program that allows them to monitor the network's activity and steal sensitive data without being detected. — rootkit, spyware
- A company's network is infected with a malicious program that modifies the boot sector of infected computers. When these computers are restarted, the program loads before the operating system, giving the hacker control over the machine. — bootkit

Example Scenarios



- A user receives a pop-up message on their computer, warning them that their system is infected with a dangerous virus and their personal data is at risk. The message urges the user to download a "free antivirus scanner" to remove the virus and protect their computer. If the user clicks on the download link, they are directed to a website that attempts to install malicious software on their computer.

  scareware

- A user downloads a free PDF reader from an untrusted website. They don't realize that the PDF reader contains malware that secretly tracks their online activity, collects their photos, and captures screenshots. The malware sends this collected data to a remote server, allowing the attackers to monitor the user's browsing habits, steal sensitive information, and potentially commit identity theft.

  trojan, spyware